Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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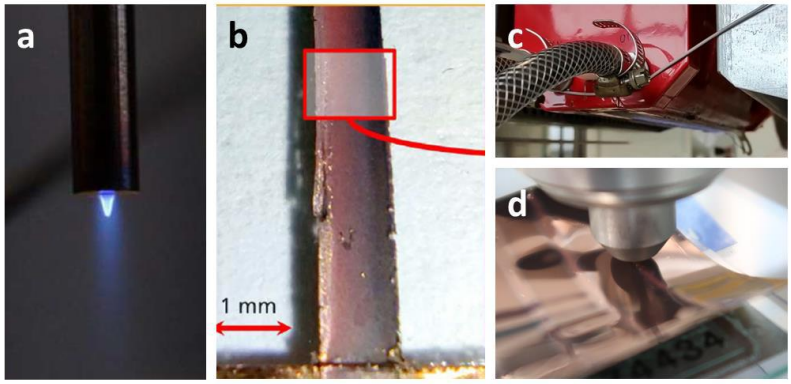
2. Technology Overview

Title of Proposed Technology	*Rapid Flame-Assisted Curing of Printed Redistribution Layers for Advanced Semiconductor Packaging Technologies*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrics & Electronic		Electrics & Electronic	
	Medium Category	Small Category	Medium Category	Small Category
	Electrical and Electronic Components	*Other Electrical and Electronic Components*	*Mechanical・Materials*	*System Integration Technologies*

Concept of Proposed Technology	• Development of a rapid RDL (Redistribution Layer) curing process for advanced packaging technologies using:
	  ○ High-resolution printing of copper (Cu) inks
	  ○ Rapid flame curing of Cu structures for post-processing (3)
	• Formation of fine-pitch Cu redistribution layers without vacuum deposition, photolithography, or electroplating.
	• Localized flame treatment provides rapid binder removal and Cu particle sintering, but is generating only a hot surface skin. Heat transfer to the glass and polymer dielectric layers is minimized.			

	 *Figure: Flame treatment using (a) a capillary flame in reductive regime with shielding gas was used to sinter (b) blade coated lines of DML inkjet Cu Ink on glass. Up-scaling of the process is envisioned using commercially available systems such as (c) a slit burner for polymer foil activation from EsseCI (2). (d) Conductive sintered lines in a full-area coated Cu layer on PI foil during measurement in digital mustimeter's measuring station.* • Enables seed-layer-free and plating-free Cu wiring with micrometer-scale line width. • Targeted for advanced semiconductor packaging, chiplet interconnection, and glass interposer platforms requiring low thermal stress and high reliability.
Rationale for Joint R&D Project	• TGV (Through-Glass Via) interposers consist of glass, Cu-filled vias, and polymer dielectric layers, which exhibit large thermal expansion mismatch and are highly sensitive to global heating during RDL post-processing. • Conventional high temperature curing and sintering processes heat up the whole wafer and induce thermo-mechanical stress. Temperature expansion differences can lead to micro-cracking of glass, interfacial delamination, and degradation of via-to-RDL reliability. • High-speed post-processing, highly localized to the wafer skin, is therefore essential for RDL formation on TGV substrates. • High-purity, oxidation-controlled Cu ink development requires advanced materials engineering and industrial-scale formulation capability. • High-resolution printing and rapid flame curing requires specialized processes in physics, equipment, and know-how not available within a single organization. • Joint R&D enables integration of Korean Cu ink material technology with Fraunhofer's high-resolution printing and flame curing process technology to establish a new RDL manufacturing platform for TGV interposers.
Key Words	• Rapid Flame Curing • Through-Glass Via (TGV) • Redistribution Layer (RDL) • High-Resolution Printing • Advanced Semiconductor Packaging

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	• Development of oxidation-protected Cu inks suitable for fine-pitch RDL printing on glass and polymer dielectric layers. • Establishment of high-resolution and high-aspect-ratio printing processes (Inkjet / EHD-jet) for micrometer-scale Cu RDL patterns. • Development of rapid flame curing processes enabling: ○ Rapid binder removal, efficient Cu particle sintering, suppression of oxidation, minimal thermal load to TGV substrates. • Demonstration of seed-layer-free and plating-free Cu RDL formation on TGV structures. • Up-scaling flame treatment from torch-flame to slit-burner. • Verification of electrical performance and thermo-mechanical reliability of printed and flame-cured RDL.			
R&D Target Specifications	• RDL line width: ≤ 20 μm (target ≤ 10 μm by EHD-jet printing) • Electrical conductivity: ≥ 10–20% of bulk Cu after flame curing • Thermal budget: No global substrate heating above polymer dielectric Tg; localized curing on hot surface skin only • Adhesion and reliability: No delamination or cracking after thermal cycling and humidity testing • Process speed: Demonstration of flame curing with 30 cm wide slit burner at ≥ 0.1 m/s for scalability			
R&D Plan	**WP1: Cu Ink Development** • Design of low-oxide, low-binder or binder-free Cu ink formulations. • Optimization of particle size, dispersion stability, viscosity, and printability for fine RDL patterns. **WP2: High-Resolution Printing of RDL** • Development of inkjet and EHD-jet printing processes on glass and polymer dielectric layers. • Control of line width, thickness, edge definition, and via-to-RDL alignment. **WP3: Rapid Flame Curing Process** • Optimization of flame conditions (gas composition, energy density, scanning speed, sheath gas). • Transfer of process parameters to slit burner for up-scale. • Minimization of thermal impact on TGV substrates while achieving sufficient Cu sintering. **WP4: Electrical and Reliability Evaluation** • Measurement of resistivity, line continuity, and contact resistance. • Thermal cycling, humidity, and adhesion tests focused on via–RDL interfaces. **WP5: Process Integration and Prototype Demonstration**			

	• Integration of printing and flame curing into a continuous RDL process flow. • Fabrication of prototype TGV interposer samples with printed Cu RDL. ✓ MS1 (Month 12): Establish oxidation-protected Cu ink and high-resolution printing (inkjet/EHD-jet) of ≤20 μm RDL lines on glass and polymer dielectric layers. ✓ MS2 (Month 24): Demonstrate rapid localized flame curing achieving Cu sintering with ≥10–20% bulk conductivity and no thermo-mechanical damage to TGV substrates. ✓ MS3 (Month 36): Validate scalable slit-burner flame curing and fabricate prototype TGV interposers with seed-layer-free, plating-free Cu RDL showing electrical and reliability compliance.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Development of high-resolution printing processes (Inkjet, EHD-jet) for Cu RDL on glass and polymer dielectrics. • Development and optimization of rapid flame curing technology for printed Cu lines. • Process physics analysis, thermal management, and oxidation control during sintering. • Electrical, structural, and chemical characterization (profilometry, conductivity, XPS, microscopy). • Demonstration of scalable curing concepts (capillary flame to slit burner).
Expected R&R of Korean SME Partner	• Development and scale-up of Cu ink formulations for RDL applications. • Control of particle synthesis, surface chemistry, dispersion stability, and rheology. • Supply of ink materials and support for printability optimization. • Participation in process integration, reliability testing, and prototype evaluation. • Preparation for industrialization and mass-production of Cu ink materials.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Establishment of a low-thermal-budget RDL formation technology for TGV glass interposers based on printed Cu and flame curing, with applicability to TSV silicon, fan-out polymer substrates, and other advanced semiconductor packaging platforms. • Demonstration of seed-layer-free and plating-free fine-pitch Cu RDL with verified electrical performance and reliability. • Creation of a new process platform combining Korean Cu ink materials and European high-resolution printing and post-curing technology, forming a generic RDL manufacturing solution for heterogeneous integration. • Reduction of process complexity and cost compared with conventional sputtering–lithography–electroplating RDL flows. • Strengthening of Korea's competitiveness in advanced semiconductor packaging, glass interposers, silicon and organic substrates, and chiplet integration technologies. • Expansion potential toward panel-level packaging and large-area heterogeneous integration markets. • Promotion of joint IP generation, standardization, and expansion into the European market based on international collaboration.

6. References

(1) DE102021131618A1

(2) Holländer, A. Adv. Flame Treat. Vakuum Forsch. Prax. **2022**, *34* (4), 43–52. https://doi.org/10.1002/vipr.202200782.

(3) https://s.fhg.de/FlamMetal

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization